

F-TALENT CODE 2026

New Fence Polygon

Time limit: 1.00 s | Memory limit: 512 MB



Problem Statement

Your grandparents have decided to build a new fence around their summer cottage this year. The cottage sits on a quiet plot of land surrounded by trees and a small garden, and the old fence has finally given up after years of wind, rain, and curious animals. They asked if you could help them get the materials, and of course you said yes. You drove to the nearest hardware store with a clear plan in mind: buy the right amount of fencing panels, posts, and screws so the project could start without delay.

Unfortunately, once you arrived at the store and stood in front of the long rows of wooden boards, you realized you had forgotten the most important piece of information — the total length of the fence. You had walked the perimeter with your grandparents the previous weekend and studied the shape carefully, but you never wrote down the actual measurements. Now you were stuck. Calling them again and again to measure every single section would be inconvenient and a bit embarrassing. You didn't want to turn a simple favor into a long series of phone calls that interrupted their afternoon.

Fortunately, you still remembered the overall shape of the fence quite well. The fence was orthogonal: every section ran either horizontally or vertically. There were no diagonal parts. Starting from one corner, the fence went in a sequence of left, right, up, and down directions, eventually returning to the exact same starting point and forming a closed loop. You could still picture the path clearly in your mind and even write it down as a string of letters — L for left, R for right, U for up, and D for down.

Each letter in that sequence represented one straight section of the fence. The lengths of those sections were unknown, but you knew they were positive real numbers. Because the path closed perfectly, the total distance traveled to the right had to equal the total distance traveled to the left, and the total distance traveled upward had to equal the total distance traveled downward. These two simple balance conditions were the only geometric constraints you needed.

You realized that if your grandparents measured every single section, you would of course know the exact perimeter. But that would require many measurements and many phone calls. The question became more interesting: what is the smallest number of sections they need to measure so that you can still calculate the precise total length of the entire fence?

In other words, you want to choose a minimal set of sections whose lengths, once known, uniquely determine the sum of all section lengths. Thanks to the two closing conditions, not every length is independent. Measuring a cleverly chosen subset can force the remaining unknown lengths into a fixed total, even if you never learn the individual values of those remaining sections.

Your task is to look at the sequence of directions and compute that minimal number. The sequence can be very long — up to a million characters — so any solution must be efficient. The answer is a single integer: the fewest measurements your grandparents need to take so you can

confidently buy exactly the right amount of fencing material and finally finish the job without further disturbance.

This small optimization turns an everyday family task into a neat computational problem. By understanding the dependencies created by the closed orthogonal path, you can reduce the number of measurements dramatically and still obtain the exact perimeter you need.

Input

The only line of the input has a string describing the shape of the fence. From the starting point, you first go some distance left ('L'), right ('R'), up ('U'), or down ('D'), then continue to the direction indicated by the second character, and so on. The path ends back at the starting point.

Output

Output the minimum number of measurements your grandparents need to take.

Constraints

The length of the string is at most 10^6 .

Example 1:

INPUT	OUTPUT
ULDRUR	2

Hint:

The fence from the example is illustrated below.

